

Swesan Pathmanathan

Canadian Citizen

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EDUCATION

McMaster University

Bachelor of Engineering, Software Engineering (Co-op)

Hamilton, ON

Sep 2021 – Apr 2026

TECHNICAL SKILLS

Languages: Python, C/C++, Verilog

Systems: Linux, Bash, Windows, ROCm, Quartus

Validation: RAS, IO, Performance, Memory, System Management

Integration: BMC, BIOS/UEFI

Data: NumPy, pandas

WORK EXPERIENCE

Advanced Micro Devices (AMD) | Python, C/C++, BMC, BIOS/UEFI, Linux, Bash, Windows, ROCm, RAS, IO, Performance, Memory, System Management

Markham, ON

Datacenter GPU System Integration Engineer (Co-op)

May 2024 – Aug 2025

- Acted as rotating **test lead** for MI300X, MI325X, and MI350X lanes, orchestrating **100+ runs weekly** on **Linux** and **Windows** while HPC and ML stress workloads ran and companion test modules exercised **RAS, IO, Performance, Memory, and System Management** under validated baselines.
- Debugged **BMC** and **BIOS/UEFI** issues on MI300X, MI325X, and MI350X systems from different customer environments, flashed and verified firmware as needed, and wrote team-facing BMC, BIOS/UEFI, and flash bring-up guides teammates reused on **20+ rack bring-ups, cutting repeat setup clarifications by about 40%** once pitfalls were centralized.
- Implemented **Python** automation against rack out-of-band interfaces for MI300X, MI325X, and MI350X with **BMC** workflows for remote power, boot, and inventory during integration bring-up, **replacing about 25 minutes of manual console work per rack with under 5 minutes** of scripted actions for the routines I automated.
- Extended **Python** automation and an internal workload launcher with version-tracked scripts, shrinking diagnostic setup from **about 20 minutes to under 5 minutes** per run because engineers reused the same canned launch paths.
- Triaged defects across **ROCm, GPU drivers, firmware, C/C++, and interconnects** using logs; added **repeatable test scripts** so every handoff shipped the same repro bundle, which **cut reopened and duplicate defects on my queue by about 30%** by eliminating one-off retests.

PROJECTS

ALU Design with Sequential Circuits | Verilog, Quartus

September 2025

- Built an **Arithmetic Logic Unit (ALU)** in **Verilog** using simple, synthesizable modules, passing all planned functional tests.
- Used **Quartus** waveform views and testbenches to check that arithmetic and logic operations behaved as expected across dozens of input cases.

Linux Log Normalization and Token Frequency Analysis | Python, NumPy, pandas

January 2026

- Developed a configurable text normalization pipeline in **Python** for processing raw **Linux logs**, supporting tokenization, stemming, lemmatization, and digit filtering.
- Performed token frequency computation and statistical aggregation using **NumPy** and **pandas** to analyze distributional patterns in **Linux logs**.

SufSort Sorting System | C, Bash, Linux

November 2025

- Wrote two sorting programs in **C** to compare how different algorithms behave on large data files of up to several hundred megabytes.
- Used **Bash** scripts on **Linux** to generate input data, run batch tests, and check sorted output, catching logic errors early in development.

RELEVANT COURSES

Computer Architecture, Real-Time Systems, Digital Systems Interfacing, Computer Performance Analysis